Natural Hazard Risk Potential

Sri Lanka



Contents

1. Geography and Demography of Sri Lanka
2. Frequent natural hazards in Sri Lanka
3. Reasons behind the high vulnerability for natuarl disasters
4. Preparedness for hazard risk mitigation
5. Move forward
6. Data availability



Geography and Demography of Sri Lanka

- Sri Lanka(SL) is an island surrounded by Indian ocean from south, Arabian sea from its west, Bay of Bengal from east; located approximately 645km north of the Equator.

- Climate - Tropical monsoons; the Northeast Monsoon (December to March), and the Southwest Monsoon (May to September) and Inter Monsoons (October-November and March -April)
- Terrain - mostly low, flat to rolling plain, with the mountainous upcountry; with 103 river basins
    - Total Area - 65,610 km^2 (Land area- 62,705 km^2, Inland water-2905 km^2)
    - Coastline-1,785 km
    - Population-22.16 million (2021)
    - Population density- 353 /km^2



Frequently available natural hazards in Sri Lanka



Floods



Lightening & Thunder storms

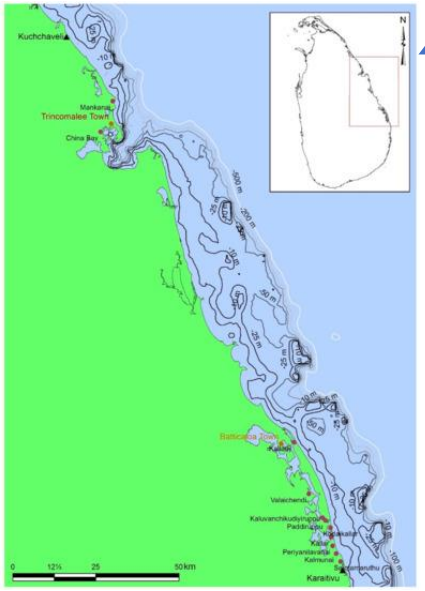


Landslides



Tsunami

FREQUENT NATURAL DISASTERS IN SL



Sea level rise



Coastal erosion



Droughts

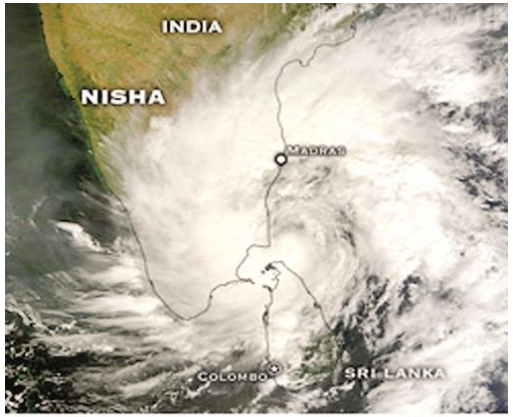
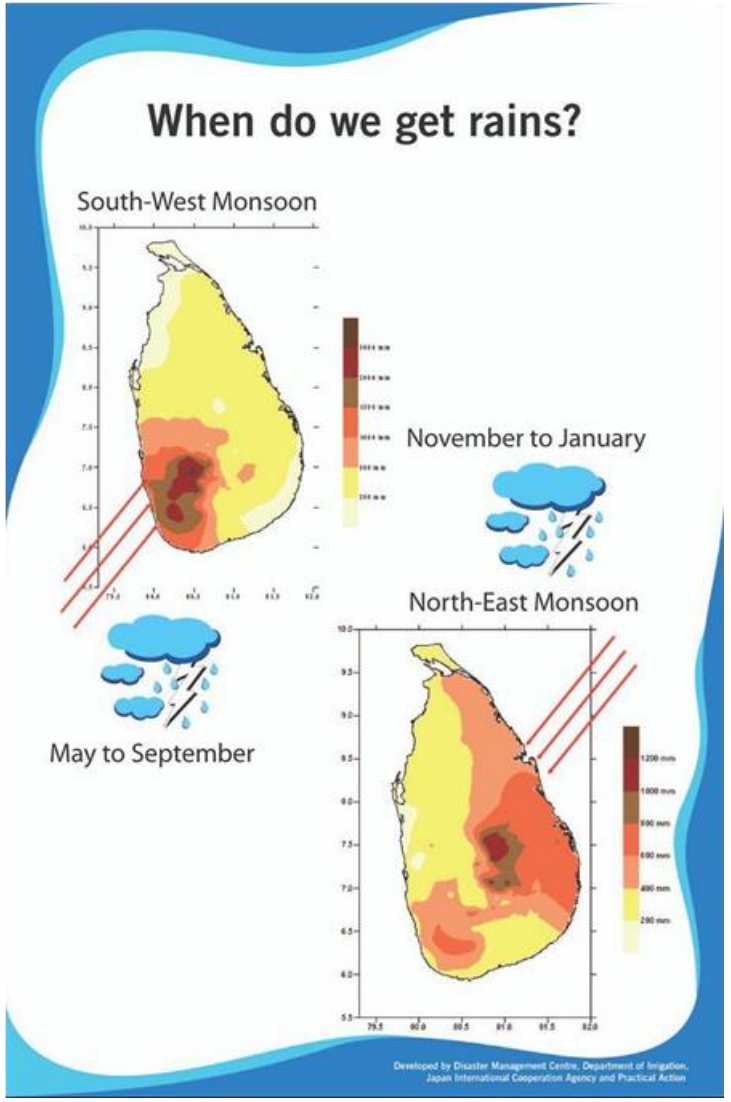


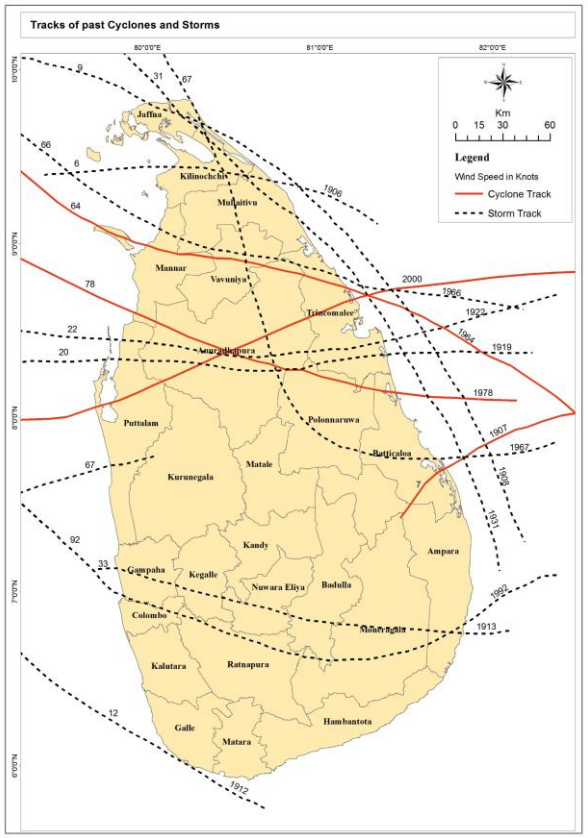
Fig. 1. Cyclone *Nisha* passing over India and Sri Lanka.

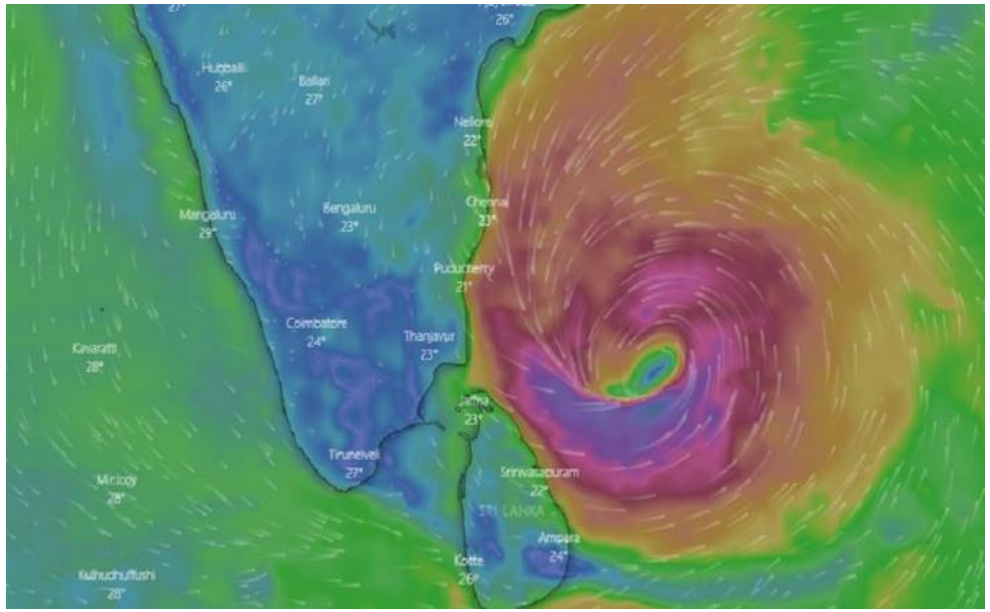
Cyclones

Reasons behind the high vulnerability for natuarl disasters

- Highly variable wind pattern around the island (wet winds and dry winds)
- Atmospheric pressure deviations in the Bay of Bengal
- Internal geography (flood plains, mountains, river network etc.)
- Deforestation
- Unplanned construction
- Climate change










U.A.E.
SAUDI-ARABIA
OMAN
EGYPT
YEMEN
SANA'A
ERITREA
DJIBOUTI
SOMALIA
Raas Xaafuun (Hafun)
ETHIOPIA
Garacad
MOGADISHU
Brava (Barawa)
KENYA
Malindi
MOMBASA
DAR ES SALAAM
TANZANIA
MOZAMBIQUE
Toamasina
ANTANANARIVO
MADAGASKAR
MAURITIUS
Réunion (F)
SEYCHELLES
VICTORIA
Mahé
Chagos Archipelago (UK)
Diego Garcia (USA)
INDIAN OCEAN
MALDIVES
MALE
SRI LANKA
COLOMBO
Beruwela
Galle
Unawatuna
Matara
Hambantota
Jaffna
Trincomalee
Batticaloa
Kerala
Pondicherry
Cuddalore
Nagapattinam
Chennai (Madras)
Bangalore
Ongole
Vishakhapatnam
Puri
Brahmapur
Kolkata (Calcutta)
Baleshwar
INDIA
BANGLADESH
Sittwe (Akyab)
MYANMAR
CHINA
VIETNAM
HANOI
LAOS
VIENTIANE
Heinan
YANGÔN (RANGOON)
THAILAND
KRUNG THEP (BANGKOK)
CAMBODIA
PHNOM PENH
Bay of Bengal
Irrawaddy-Delta
Precanis
Coco-Islands
Dawei
Mergui-Archipelago
Port Blair
ANDAMAN-SEA
Kawthaung
Gulf of Thailand
Khao Lak
Phuket
Phang Nga
Krabi
Koh Phi Phi
Koh Lanta
Ko Lanta
Ko Lanta Yai
Ko Lanta Noi
Ko Lanta Kao
Ko Lanta Nua
Ko Lanta Tai
Ko Lanta Klang
Ko Lanta Nok
Ko Lanta Nai

National preparedness for disasters

- In May 2005, the government of Sri Lanka enacted the Disaster Management Act No.13 of 2005.
  - This provides the legal basis for a DRM system in the country.
  - The Disaster Management Centre (DMC) was established to implement the functions indicated in the Act.
  - A National Disaster Management Plan was constructed:
- National Disaster Management Plan
  - Risk Assessment
  - Disaster Preparedness Plan for Emergency Response
  - Disaster Prevention and Mitigation Plan
  - Contingency Plans
  - Disaster Rehabilitation and Reconstruction Plan

Move forward

- The available disaster risk management plan is at a satisfactory level in terms of risk assessment, preparedness and emergency response.
- However, there is a need to fill in the data and resource gaps in some areas with regard to hazard risk vulnerability (which was resulted due to financial constratints in the recent years).
- It is important to address the financial risk associated with the natural hazards in which not yet taken into consideration in the risk management plan.
- There is a need for a long term planned mechanism in order to minimize the risk of natural hazards in Sri Lanka.

Data availability

Disaster management centre, Sri Lanka
http://www.dmc.gov.lk

Ministry of Disaster Management,Sri Lanka
https://disastermin.gov.lk

Department of Census and Statistics, Sri Lanka
http://www.statistics.gov.lk

Department of Meteorology, Sri Lanka
https://www.meteo.gov.lk

National building research organisation
https://nbro.gov.lk